

MODEL

Product
- id: String - name: String - description: String - price: double - costPrice: double - quantity: int - lowStockThreshold: int - categoryId: String - supplierId: String - dateAdded: LocalDate
+ isLowStock(): boolean + getProfitMargin(): double + getStockValue: double

Category
- id: String - name: String - description: String

Supplier
- id: String - name: String - contactName: String - phone: String - email: String - address: String

DAO LAYER

(interface)

ProductDAO
+ save(p: Product) + update(p: Product) + delete(id: String) + findById(id: String) + findAll(): List<Product> + searchByName(name: String) + findLowStock()

ProductDAOImpl
- filePath: String
+ save(p: Product) + update(p: Product) + delete(id: String) + findAll(): List<Product> - loadAll(): List<Product>

CategoryDAO	(interface)
+ save(c: Category) + update(c: Category) + delete(id: String) + findAll(): List<Category>	

CategoryDAOImpl
- filePath: String
+ loadAll(): List<Category>

(interface) SupplierDAO
+ save (s: Supplier) + update (s: Supplier) + delete (id: String) + findAll(): List<Supplier>

SupplierDAOImpl
- filepath: String
- loadAll(): List<Supplier>

SERVICE LAYER

ProductService
- productDAO: ProductDAO
+ addProduct (p: Product) + updateProduct (p: Product) + deleteProduct (id: String) + getAllProducts(): List<Product> + searchByName (name: String) + getLowStockProducts()

CategoryService
- categoryDAO: CategoryDAO
+ addCategory (c: Category) + updateCategory (c: Category) + deleteCategory (id: String) + getAllCategories(): List

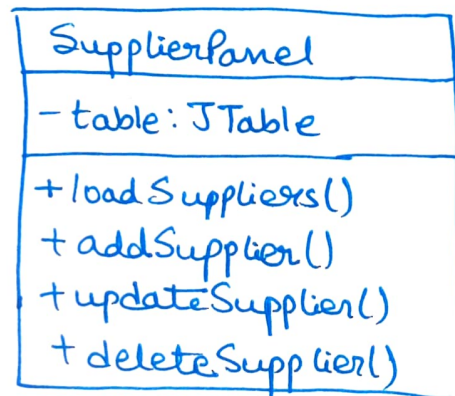
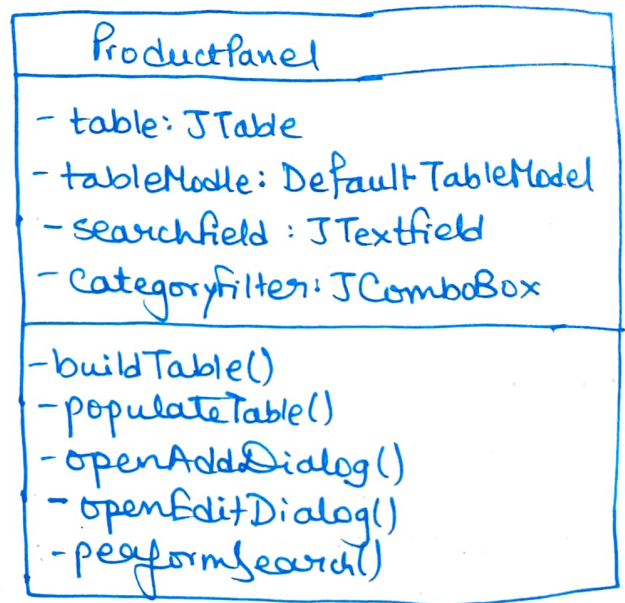
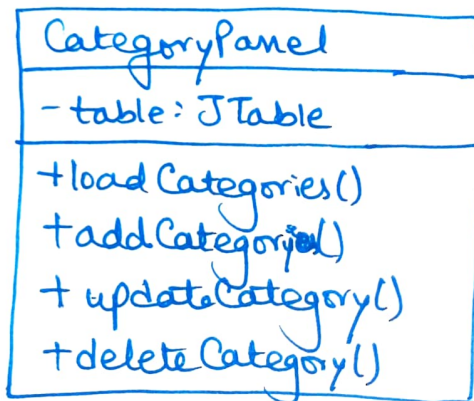
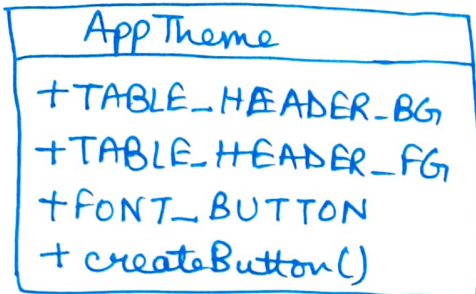
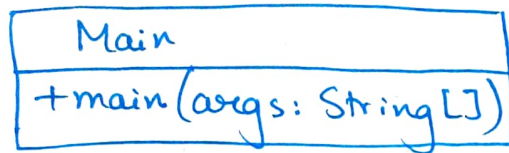
SupplierService
- supplierDAO: SupplierDAO
+ addSupplier (s: Supplier) + updateSupplier (s: Supplier) + deleteSupplier (id: String) + getAllSuppliers(): List

UTILITY

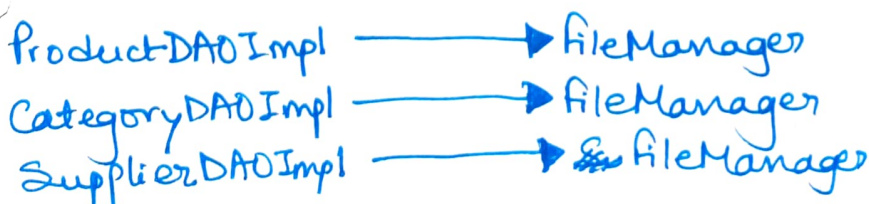
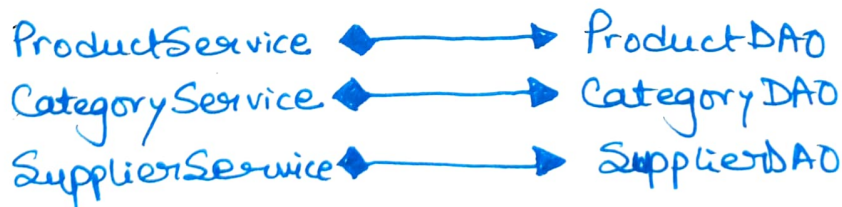
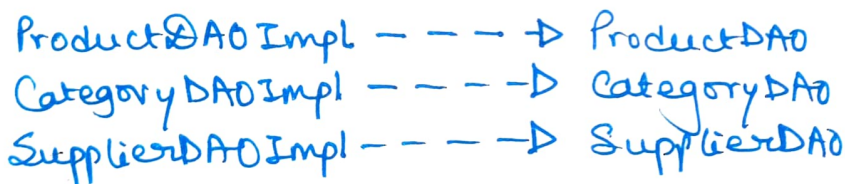
FileManager
+ saveData (list, filePath) + loadData (filePath)

CSVExporter
+ exportProducts(list)

UI (SWING)



RELATIONSHIPS



ProductPanel → ProductService
CategoryPanel → CategoryService
SupplierPanel → SupplierService

Main → ProductPanel
Main → CategoryPanel
Main → SupplierPanel

Product → Category
Product → Supplier

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